

AC4101 — Management Accounting: Performance and Decision Making

Summer 2022 · Paper 1 · Full Marking Scheme

Module Code	AC4101
Module Title	Management Accounting: Performance and Decision Making
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Question	Scenario	Marks
Question 1	FloClean Ltd — Traditional Absorption Costing vs Activity Based Costing	50
Question 2	Wavewise Ltd — DCF Comparison: Purchase vs Rent, Payback & CSR	50
Question 3	Bright Ltd — Balanced Scorecard & Non-Financial Objectives	50

Instructions to Candidates (as stated on the paper): Answer ANY TWO questions. Show all workings clearly – marks will be awarded for workings.

Note on this marking scheme: Numerical parts are marked primarily on correct method and workings, with the **own-figure rule** applied throughout. Discursive parts are marked using qualitative bands assessing depth of analysis, application to the specific scenario, and clear structure. No academic citations are expected or required in any closed-book exam answer.

QUESTION 1

FloClean Ltd

Total: 50 marks

This question compares Traditional Absorption Costing with Activity Based Costing for a water filter manufacturer producing three product types. It tests overhead absorption mechanics, ABC cost driver rate calculation, profit statement preparation under both systems, and discursive understanding of over-absorption consequences and ABC benefits.

Parts in this Question

Part	Requirement	Marks
(a)	Prepare a Profit Statement using Traditional Absorption Costing	10
(b)	Calculate the cost driver rates under the proposed ABC method	8
(c)	Prepare a Profit Statement using Activity Based Costing	15
(d)	Comment on reasons for any difference in costs under each system	5
(e)	TWO potential negative consequences of over-absorbing overheads	6
(f)	TWO benefits of using an Activity Based Costing system	6

Question 1(a)**10 marks**

Prepare and present a Profit Statement using traditional absorption costing.

Full Model Answer

Overhead absorbed per unit – Machine Department at €1.80/machine hour + Assembly Department at €1.30/direct labour hour:

	Tap	Sink	Delux
Machine OH (hrs/unit × €1.80)	2 × 1.80 = 3.60	4 × 1.80 = 7.20	7 × 1.80 = 12.60
Assembly OH (hrs/unit × €1.30)	3 × 1.30 = 3.90	4 × 1.30 = 5.20	4 × 1.30 = 5.20
Total OH absorbed per unit	€7.50	€12.40	€17.80

	Tap (€)	Sink (€)	Delux (€)	Total (€)
Units	14,000	16,500	18,000	–
Sales	4,200,000	9,075,000	12,420,000	25,695,000
Direct materials	1,190,000	2,227,500	4,680,000	8,097,500
Direct labour	672,000	1,188,000	1,728,000	3,588,000
Production overhead absorbed	105,000	204,600	320,400	630,000
Total cost	1,967,000	3,620,100	6,728,400	12,315,500
Profit	2,233,000	5,454,900	5,691,600	13,379,500

Reconciliation note: Total OH absorbed under traditional costing = €630,000, which is significantly less than the total overhead pool of €759,200. This represents an under-absorption of €129,200, consistent with the pre-set absorption rates being based on a wider company-wide activity base than just these three products. This is relevant to part (d)'s comparison discussion.

Mark Allocation

Marks	Criteria
9-10	Correct OH per unit for all three products using both Machine (€1.80/mhr) and Assembly (€1.30/DLH) rates; correct total profit of €13,379,500; clear layout with workings shown.
7-8	Correct method throughout with one minor error (e.g. one product's overhead absorption); own-figure rule applied; layout clear.
5-6	Correct overall structure but errors in two or more overhead calculations, or one absorption rate misapplied (e.g. machine rate used for assembly hours).
2-4	Some correct elements (sales, direct costs) but overhead absorption method substantially wrong or omitted.
0-1	No meaningful attempt.

Question 1(b)**8 marks**

Calculate the cost drivers under the proposed Activity Based Costing method.

Full Model Answer

Each cost driver rate = Cost Pool ÷ Total Quantity for Period:

Cost Pool	Cost (€)	Cost Driver	Qty for Period	Rate
Machining services	225,000	Machine hours	220,000	€1.0227/mhr
Assembly services	300,000	Direct labour hours	180,000	€1.6667/DLH
Set-up Costs	35,200	Number of set-ups	1,100	€32.00/set-up
Order processing	104,000	Customer orders	32,000	€3.25/order
Purchasing	95,000	Supplier orders	11,377	€8.3502/order

Note: The “quantity for period” figures represent total company-wide activity, not just the sum of these three products’ usage. This is why ABC overhead recovered across only these three products will be less than the full €759,200 pool – the same principle as the under-absorption seen in part (a).

Mark Allocation

Marks	Criteria
7-8	All five cost driver rates correctly calculated (cost ÷ quantity for period) with workings; correct identification of each driver’s cost pool.
5-6	Four of five rates correct, or correct method throughout with minor arithmetic slips.
3-4	Correct method shown for most pools but with mismatched driver/pool pairings or significant arithmetic errors.
1-2	Some understanding shown but rates largely incorrect or product-level figures used instead of period totals.
0	No meaningful attempt.

Question 1(c)**15 marks**

Using your results from part (b), prepare and present a Profit Statement using Activity Based Costing.

Full Model Answer**Step 1: Total activity consumption by product**

	Tap	Sink	Delux
Machine hours (units × hrs/unit)	14,000×2 = 28,000	16,500×4 = 66,000	18,000×7 = 126,000
Direct labour hours (units × hrs/unit)	14,000×3 = 42,000	16,500×4 = 66,000	18,000×4 = 72,000
Number of set-ups	420	300	380
Customer orders	7,500	8,400	16,100
Supplier orders	3,100	4,050	4,200

Step 2: Overhead cost by activity pool (usage × rate from part (b))

	Tap (€)	Sink (€)	Delux (€)
Machining (€1.0227/mhr)	28,636	67,500	128,864
Assembly (€1.6667/DLH)	70,000	110,000	120,000
Set-up (€32.00/set-up)	13,440	9,600	12,160
Order processing (€3.25/order)	24,375	27,300	52,325
Purchasing (€8.3502/order)	25,886	33,818	35,071
Total ABC overhead	162,337	248,218	348,420

Step 3: ABC Profit Statement

	Tap (€)	Sink (€)	Delux (€)	Total (€)
Sales	4,200,000	9,075,000	12,420,000	25,695,000
Direct materials	1,190,000	2,227,500	4,680,000	8,097,500
Direct labour	672,000	1,188,000	1,728,000	3,588,000
ABC overhead	162,337	248,218	348,420	758,975
Total cost	2,024,337	3,663,718	6,756,420	12,444,475
Profit	2,175,663	5,411,282	5,663,580	13,250,525

Reconciliation: Total ABC overhead recovered (€758,975) is very close to the full €759,200 pool – the minor €225 difference is a rounding effect from the Purchasing rate ($95,000 \div 11,377 = €8.3502\dots$). All three products are less profitable under ABC than under traditional costing, as ABC distributes the full overhead pool more accurately across each product's actual consumption of each activity.

Mark Allocation

Marks	Criteria
13-15	Full ABC profit statement with correct activity consumption (Step 1), correct overhead cost by pool (Step 2), and correct profit figures; own-figure rule applied from part (b) rates; total profit approximately €13,250,525.
10-12	Correct method with minor errors in one or two activity cost calculations; clear workings; own-figure rule applied.
6-9	Correct ABC structure but errors across multiple cost pools, or activity consumption figures (Step 1) partly miscalculated.
2-5	Partial ABC approach; some pools correctly applied but others absorbed on a volume basis or omitted.
0-1	No meaningful ABC method applied.

Question 1(d)**5 marks**

Comment on the reasons for any difference in the costs calculated under each system.

Full Model Answer**Under Traditional Costing, Overheads Are Absorbed Using Volume-Based Drivers Only**

The traditional system allocates production overheads using only machine hours and direct labour hours – two activity measures that are directly proportional to production volume. This means all three products are charged overhead in simple proportion to how many hours they consume in the Machine and Assembly departments, ignoring the fact that some products generate far more set-up activity, customer order transactions, or purchasing complexity than others.

ABC Recognises the True Drivers of Each Overhead Pool

Under ABC, each cost pool is driven by the activity that actually causes it. For example, the Delux filter generates 16,100 customer orders (compared to 7,500 for Tap) – under traditional costing this transactional intensity is completely invisible. ABC charges Delux more order-processing overhead (€52,325) than Tap (€24,375), accurately reflecting its higher consumption of this activity. Similarly, set-up costs are driven by the number of set-ups rather than volume, which shifts costs differently across all three products.

All Products Are Less Profitable Under ABC

Because traditional costing under-absorbs €129,200 of the total €759,200 overhead pool (only recovering €630,000), while ABC recovers nearly the full €759,200, all three products bear higher total overhead under ABC, resulting in lower profits across the board. The key management insight is that traditional costing was understating the true cost of all three products.

Mark Allocation

Marks	Criteria
5	Clear explanation of why results differ: traditional uses only volume-based drivers (machine/DLH hours) vs ABC using activity-specific drivers; reference to specific numerical examples from the question (e.g. Delux's high customer order volume); explanation of the under-absorption difference.
3-4	Correct explanation of the conceptual difference between traditional and ABC with some numerical reference; most of the key points covered.
2	General statement that ABC is more accurate but without specific explanation of why the numbers differ in this scenario.
1	Vague reference to overhead allocation differences without meaningful explanation.
0	No meaningful attempt.

Question 1(e)**6 marks**

Advise the management team of FloClean of TWO potential negative consequences of over-absorbing overheads and explain how this situation could be avoided.

Full Model Answer**Overstated Unit Costs Leading to Non-Competitive Pricing**

Over-absorption occurs when the pre-set overhead rate recovers more overhead than was actually incurred. Each unit of Tap, Sink, or Delux filter will carry an inflated overhead charge, making their standard costs appear higher than they truly are. If FloClean's management team uses these inflated costs to set selling prices, the products may be priced above the competitive market level, potentially causing FloClean to lose sales to lower-priced competitors in the domestic water filter market.

Distorted Product Profitability Signals

Over-absorbed overhead suppresses reported per-unit margins during the period, potentially leading management to incorrectly identify profitable product lines as marginal or unprofitable. For FloClean, which distributes through wholesalers, inaccurate product profitability signals could trigger poor product mix decisions – for example, diverting resources away from the Delux filter if it appears less profitable than it truly is after the year-end over-absorption adjustment.

How This Can Be Avoided

FloClean should regularly review and update its predetermined overhead absorption rates – ideally on a rolling quarterly basis – to ensure they reflect current actual activity levels rather than outdated budgeted figures. Adopting ABC, as proposed in this question, also reduces the risk of systematic over- or under-absorption because it ties recovery directly to the actual activity levels driving each overhead pool.

Mark Allocation

Marks	Criteria
5-6	Two distinct negative consequences clearly explained and applied to FloClean's context; practical avoidance method described.
3-4	Two consequences identified with reasonable explanation; avoidance method present but generic.
2	One consequence properly explained; avoidance method underdeveloped.
1	Vague reference to over-absorption being problematic without specific consequences named.
0	No meaningful attempt.

Question 1(f)**6 marks**

Explain TWO benefits to FloClean of using an Activity Based Costing system.

Full Model Answer**More Accurate Product Costing Supports Better Pricing and Mix Decisions**

As demonstrated in part (c), ABC reveals that all three products carry more overhead than the traditional system suggested – particularly because traditional costing was only absorbing €630,000 of a €759,200 pool. ABC's more accurate cost allocation enables FloClean's management to set prices that genuinely cover all overhead, and to make informed product mix decisions. For example, if ABC showed the Delux filter was less profitable than expected due to its high customer order activity, management could reassess Delux's pricing or reduce order complexity to improve margins.

Improved Overhead Cost Control Through Activity Visibility

By separating overheads into five distinct activity pools (machining, assembly, set-ups, order processing, purchasing), ABC gives FloClean's management clear visibility of which activities are driving overhead spend. This enables targeted cost-reduction initiatives – for instance, if set-up costs (€35,200) are high, management can focus on batch-size optimisation to reduce set-up frequency. Under the traditional two-rate system, this level of cost driver insight is simply not available.

Mark Allocation

Marks	Criteria
5-6	Two distinct benefits clearly explained and applied to FloClean's specific context, ideally referencing the data (e.g. the under-absorption gap, specific cost pools).
3-4	Two benefits identified with reasonable explanation; limited application to FloClean specifically.
2	One benefit properly explained, or both identified with minimal explanation.
1	Vague reference to ABC being "more accurate" without explanation.
0	No meaningful attempt.

QUESTION 1 TOTAL — 50 MARKS

QUESTION 2

Wavewise Ltd

Total: 50 marks

This question tests discounted cash flow investment appraisal comparing two equipment options (purchase vs rent) for a wave energy data analytics company. It covers NPV for both alternatives, payback period for the purchase option, an investment recommendation, treatment of the existing equipment's resale value, and CSR policy development.

Parts in this Question

Part	Requirement	Marks
(a)	Evaluate both alternatives using discounted cash flow (NPV for Equipment A & B)	20
(b)	Calculate the payback period for Equipment A (purchase option)	5
(c)	Advise which alternative to recommend using results from (a) and (b)	10
(d)	Explain treatment of the €20,000 current cash value of existing equipment	5
(e)	THREE CSR goals (environmental and social) with practices to achieve them	10

Question 2(a)**20 marks**

Evaluate the two alternatives using discounted cash flow techniques.

Full Model Answer

Key principles applied: (1) Depreciation is a non-cash accounting charge and is **excluded** from all DCF calculations. (2) The existing equipment can be sold now for €20,000 – this is a Year 0 cash inflow included under both options. (3) The €20,000 resale value offsets the Year 0 capital outlay under Equipment A; under Equipment B there is no capital purchase so the €20,000 is a straight Year 0 inflow. (4) 10% cost of capital, 3-year project life for both options.

Equipment A – Purchase Option

	Calculation	Amount (€)
Year 0: Capital cost	–	(285,000)
Year 0: Sale of existing equipment	–	20,000
Net Year 0 outflow	–	(265,000)
Annual net cash flow: savings less fixed costs	600,000 – 80,000	520,000 per year

Year	Net CF (€)	Discount Factor (10%)	Present Value (€)
1	520,000	0.909	472,680
2	520,000	0.826	429,520
3	520,000	0.751	390,520
Total PV of cash inflows			1,292,720
Less: Net Year 0 outflow			(265,000)
NPV – Equipment A (Purchase)			€1,027,720

Equipment B – Rent Option

	Calculation	Amount (€)
Year 0: Sale of existing equipment (no capital purchase)	–	20,000
Annual net cash flow: savings less fixed costs and rental	230,000 – 65,000 – 200,000	(35,000) per year

Year	Net CF (€)	Discount Factor (10%)	Present Value (€)
1	(35,000)	0.909	(31,815)
2	(35,000)	0.826	(28,910)
3	(35,000)	0.751	(26,285)
Total PV of cash flows			(87,010)
Plus: Year 0 inflow (sale of existing)			20,000
NPV – Equipment B (Rent)			€(67,010)

Mark Allocation

Marks	Criteria
18–20	Both NPVs correctly calculated; depreciation correctly excluded; €20,000 existing equipment proceeds included at Year 0 in both; correct annual net CFs (A: €520,000; B: €–35,000); correct discounting; Equipment A NPV €1,027,720; Equipment B NPV €(67,010).
14–17	Both options attempted with correct structure; one NPV correct and one with minor errors; depreciation correctly excluded; own-figure rule applied.
10–13	Correct DCF structure for both options; errors in one or more cash flow components (e.g. depreciation included, or existing equipment proceeds omitted); discounting correctly applied.
5–9	One option substantially correct; second option attempted but with significant errors; or both options have correct structure but multiple cash flow errors.
0–4	No meaningful DCF attempt, or fundamental misunderstanding of NPV mechanics.

Question 2(b)**5 marks**

For the alternative to purchase Equipment A, calculate the payback period.

Full Model Answer

Year	Net Cash Flow (€)	Cumulative (€)
0	(265,000)	(265,000)
1	520,000	255,000

Payback occurs during Year 1 – the cumulative cash flow turns positive after just one year's net inflow.

Amount to recover at start of Year 1	€265,000
Year 1 net cash flow	€520,000
Fraction of Year 1 required	$265,000 \div 520,000 = 0.5096$
Payback Period	0.5096 years $\approx$ 6.1 months

Note: The very short payback of approximately 6 months reflects the exceptionally high annual savings of €600,000 relative to the net capital outlay of €265,000. This is strong supporting evidence for the recommendation in part (c).

Mark Allocation

Marks	Criteria
5	Correct net Year 0 outflow of €265,000 used (not gross €285,000); correct identification that payback falls within Year 1; correct fractional calculation giving approx 0.51 years / 6.1 months.
3-4	Correct method; minor error e.g. using €285,000 instead of €265,000 as starting point (own-figure from part (a) accepted); fractional year correctly calculated.
2	Correct payback year identified; fractional calculation omitted or minor error.
1	Cumulative cash flow approach attempted but significant errors.
0	No meaningful attempt.

Question 2(c)**10 marks**

Advise the management of Wavewise Ltd which alternative you recommend, using the results from your calculations above to explain your answer.

Full Model Answer**Clear Recommendation: Equipment A (Purchase) Is Strongly Preferred**

Equipment A generates an NPV of €1,027,720, representing a substantial positive return on investment above Wavewise Ltd's 10% cost of capital hurdle rate. Equipment B (Rent) generates a deeply negative NPV of €(67,010), meaning it destroys value rather than creating it. On both NPV and payback criteria, Equipment A is decisively superior and should be selected.

Payback Confirms the Strength of the Purchase Case

Equipment A's payback period of approximately 6.1 months is exceptionally fast, meaning Wavewise Ltd recovers the net capital outlay within the first half of Year 1. This rapid payback significantly reduces the financial risk of the investment and indicates a very healthy level of annual cash generation (€520,000 net per year) relative to the capital commitment (€265,000 net of the existing equipment sale proceeds).

Why Equipment B Fails Financially

Under Equipment B, the annual rental cost of €200,000 per year effectively consumes almost all of the €230,000 annual savings, leaving only a €35,000 per year net inflow before fixed costs – which actually produces a negative net annual cash flow of €(35,000). The rental model transfers most of the value created by the improved equipment to the equipment owner rather than to Wavewise Ltd. Management should also consider that Equipment B provides no terminal ownership asset, whereas Equipment A's purchase creates a capital asset with potential residual value beyond the 3-year horizon.

Mark Allocation

Marks	Criteria
9-10	Clear recommendation for Equipment A; direct reference to NPV figures (€1,027,720 vs €(67,010)) and payback (6.1 months); explanation of why Equipment B fails (rental consumes savings); own-figure rule applied throughout.
7-8	Clear recommendation for Equipment A with correct use of NPV comparison; payback mentioned; one or more supporting arguments for the recommendation.
4-6	Correct recommendation but loosely justified; limited quantitative reference to the calculation results.
1-3	Recommendation stated but without meaningful financial justification from parts (a)/(b).
0	No meaningful attempt.

Question 2(d)**5 marks**

Explain your treatment of the €20,000 current cash value of the existing equipment.

Full Model Answer**The €20,000 Is a Relevant Incremental Cash Flow - Not a Sunk Cost**

The €20,000 is the current market (resale) value of the existing scanning equipment. Unlike the original purchase cost of the existing equipment (which is a sunk cost – already spent, irrecoverable), the €20,000 represents a cash flow that Wavewise Ltd *will actually receive* if it proceeds with either replacement option. Because the decision to replace triggers this cash receipt, it is a **relevant, incremental cash flow** and must be included in the NPV analysis.

Timing and Treatment in the NPV

The €20,000 is treated as a **Year 0 cash inflow** under both Equipment A and Equipment B, since the existing equipment would be sold at the point of replacement (January 2024, i.e. the start of the project). Under Equipment A it reduces the net capital outlay from €285,000 to €265,000. Under Equipment B, where there is no capital purchase, it is simply a Year 0 inflow. Including it in both options ensures a consistent and fair comparison between the two alternatives.

Mark Allocation

Marks	Criteria
5	Clear explanation that €20,000 is a relevant incremental cash flow (not a sunk cost); correctly identified as a Year 0 inflow; correctly applied to both Equipment A (reducing net outlay) and Equipment B; explanation of why it is included (it arises as a result of the replacement decision).
3–4	Correct identification as relevant/incremental; Year 0 timing correct; treatment in one option clearly explained.
2	Correct identification that it should be included but limited explanation of why or how it is applied.
1	Some awareness that the €20,000 matters to the decision but incorrect or missing explanation.
0	No meaningful attempt.

Question 2(e)**10 marks**

The general manager of Wavewise Ltd wishes to develop a strong Corporate Social Responsibility (CSR) policy both from an environmental and a social perspective. Advise her of THREE goals that should be formally agreed in a CSR policy, outlining practices needed to achieve these goals.

Full Model Answer**Environmental Goal 1: Minimise Carbon Footprint of Wave Energy Data Operations**

Goal: Achieve net-zero operational carbon emissions from Wavewise Ltd's data gathering and analytics activities by 2030, aligned with the company's positioning in the renewable wave energy sector. Practices: Power all data centre and office operations from renewable energy sources; measure and publish annual Scope 1 and Scope 2 carbon emissions; establish a carbon offset programme for unavoidable emissions; replace diesel-powered field equipment with battery or hydrogen alternatives where feasible; report progress against targets in an annual sustainability report.

Environmental Goal 2: Responsible Management of Scanning Equipment and E-Waste

Goal: Ensure all replaced scanning equipment (such as the equipment being replaced in this decision) is responsibly recycled or repurposed rather than sent to landfill. Practices: Partner with certified electronics recycling providers for end-of-life equipment disposal; prioritise refurbishment and donation of operational equipment to educational or research institutions; establish a supplier take-back policy requiring new equipment vendors (such as the supplier of Equipment A) to accept end-of-life units for responsible recycling; track and report on the tonnage of equipment diverted from landfill annually.

Social Goal: Support STEM Education and Community Engagement in Coastal Communities

Goal: Contribute to the education and skills development of communities in coastal areas where Wavewise Ltd operates its wave energy monitoring activities. Practices: Establish an annual scholarship or bursary fund for students studying marine science, data analytics, or renewable energy at third-level institutions; provide work placement opportunities for students from local communities each year; partner with local secondary schools to deliver STEM outreach programmes focused on wave energy and data science; measure impact through the number of beneficiaries supported annually and report in the CSR policy review.

Mark Allocation

Marks	Criteria
9-10	Three distinct goals covering both environmental and social dimensions as requested; each goal is specific, measurable, and applied to Wavewise Ltd's wave energy/data analytics context; concrete, achievable practices included for each goal.
7-8	Three goals spanning both dimensions; relevant to context; practices included but less detailed or one goal is generic rather than industry-specific.
4-6	Two well-developed goals with practices, or three goals identified with minimal practices; both environmental and social aspects may not be covered.
1-3	One or two goals identified; practices vague or absent; goals entirely generic with no application to Wavewise Ltd.
0	No meaningful attempt.

QUESTION 2 TOTAL — 50 MARKS

QUESTION 3

Bright Ltd & Non-Financial Objectives

Total: 50 marks

This question has two parts. Part 1 tests the Balanced Scorecard framework in the context of Bright Ltd, an Irish mobile operator exploring international expansion – covering all four BSC perspectives with objectives and measures, and the link between BSC and strategy/financial outcomes. Part 2 is a standalone essay-style question on non-financial objectives and their stakeholder groups.

Parts in this Question

Part	Requirement	Marks
Part 1(a)	Explain the Balanced Scorecard; include one performance measure for each of the four perspectives for Bright Ltd, with a reason to support each measure	20
Part 1(b)	Explain how the Balanced Scorecard links strategy formulation to financial outcomes	15
Part 2	THREE non-financial objectives organisations might have, identifying the stakeholder group for each	15

Question 3 · Part 1(a)**20 marks**

Explain the Balanced Scorecard and how it could be used by Bright Ltd to measure its current performance. Include one performance measure that is appropriate for Bright Ltd for each of the four perspectives, giving a reason to support each measure.

Full Model Answer**What the Balanced Scorecard Is**

The Balanced Scorecard (BSC), developed by Kaplan and Norton, is a strategic performance management framework that translates an organisation's strategy into a balanced set of objectives and measures across four perspectives: Financial, Customer, Internal Business Process, and Learning & Growth. Unlike traditional performance measurement systems which focus almost entirely on financial metrics (revenue, profit, cost), the BSC recognises that financial outcomes are the *result* of non-financial activities and that managing only financial outcomes is backward-looking. The BSC gives managers a more complete and forward-looking picture of organisational health.

Financial Perspective — Measure: Revenue Growth Rate

This measures the percentage increase in total revenue year-on-year. **Reason:** Bright Ltd is currently assessing international expansion – revenue growth rate directly captures whether the core domestic business is growing sufficiently to fund expansion, and will be the primary metric investors and the board use to evaluate whether international expansion is generating returns. It is the most fundamental indicator of commercial success for a mobile operator.

Customer Perspective — Measure: Customer Churn Rate (or Net Promoter Score)

Customer churn rate measures the percentage of customers who cancel their mobile subscription in a given period. **Reason:** In the highly competitive Irish mobile market, retaining existing customers is critical and less costly than acquiring new ones. A rising churn rate signals deteriorating customer satisfaction, quality of service, or price competitiveness – all of which would undermine Bright Ltd's ability to fund international expansion from a stable domestic revenue base. Monitoring churn directly measures customer loyalty, which is fundamental to sustainable growth.

Internal Business Process Perspective — Measure: Network Uptime / Service Availability (%)

This measures the percentage of time the mobile network operates without outage. **Reason:** For a mobile operator, network reliability is the core product. Customers' willingness to stay with Bright Ltd (and recommend it to others) depends primarily on network quality. High network uptime is also a prerequisite for international expansion – demonstrating operational excellence in Ireland builds the internal process capability that can be replicated in new markets. This measure directly reflects the efficiency and effectiveness of Bright Ltd's core internal processes.

Learning & Growth Perspective — Measure: Employee Training Hours per Staff Member per Year

This measures the average number of hours of training and development completed per employee annually. **Reason:** Successfully entering international markets requires Bright Ltd's workforce to develop new capabilities – in market entry strategy, regulatory compliance in new jurisdictions, and cross-cultural customer service. Tracking training hours ensures the organisation is investing in the human capital necessary to execute the international expansion strategy. Without skilled, adaptable people, strategy formulation and network excellence cannot be sustained.

Mark Allocation

Marks	Criteria
18-20	Clear, accurate explanation of the BSC framework (4 perspectives, balancing financial and non-financial); one appropriate, specific measure for each perspective clearly applied to Bright Ltd's mobile operator context; a genuine, well-reasoned justification for each measure's relevance to Bright Ltd (not just a generic description of the perspective).
14-17	Good explanation of BSC; four measures identified, at least three with strong, applied justifications; one measure may be generic or the justification underdeveloped.
10-13	Reasonable BSC explanation; four perspectives covered; measures mostly correct but justifications weak or not specifically applied to Bright Ltd's context.
5-9	BSC concept partially understood; fewer than four perspectives covered, or measures given without meaningful justification.
0-4	No meaningful understanding of the BSC demonstrated.

Question 3 · Part 1(b)**15 marks**

Explain how the Balanced Scorecard links strategy formulation to financial outcomes.

Full Model Answer**The BSC as a Strategy Execution Tool**

The Balanced Scorecard is not just a measurement system – it is a strategic management tool. Its fundamental purpose is to translate high-level strategy into specific, measurable actions across the organisation, and to trace a logical path from those actions to financial outcomes. Kaplan and Norton described this as a “strategy map”: a visual representation of the cause-and-effect chain from Learning & Growth (people and systems) through Internal Processes (what we do well) to Customer outcomes (satisfaction, loyalty) to Financial results (revenue, profit).

The Cause-and-Effect Chain: From Learning to Financial Results

For Bright Ltd, the linkage works as follows: investing in employee training and development (Learning & Growth) improves the quality of network management and customer service processes (Internal Business Process). Better processes lead to fewer network outages, faster fault resolution, and a smoother customer experience, which increases customer satisfaction and reduces churn (Customer perspective). Satisfied, loyal customers generate stable recurring subscription revenue and are more likely to upgrade plans or recommend Bright Ltd to others, which drives revenue growth and improved operating margins (Financial perspective). Every non-financial objective in the BSC ultimately traces back to a financial consequence.

Strategy Formulation: Translating Vision Into Objectives

Strategy formulation begins with the organisation’s vision and strategic priorities. For Bright Ltd, the managing director’s goal of international expansion would be embedded in the BSC as strategic objectives across all four perspectives – for example: “Develop international market entry capability” (Learning & Growth), “Build scalable network infrastructure” (Internal Process), “Acquire and retain customers in new markets” (Customer), and “Generate return on international investment above cost of capital” (Financial). The BSC ensures these are not just aspirations but are translated into specific, measurable targets with accountability assigned.

Avoiding Short-Termism

A critical benefit of the BSC link between strategy and financial outcomes is that it discourages management from focusing exclusively on short-term financial targets (e.g. cutting training budgets to boost this year’s profit) at the expense of long-term strategic capability. By tracking non-financial leading indicators alongside financial lagging indicators, the BSC makes it visible when short-term financial improvements are being achieved at the cost of eroding the capabilities that drive future financial performance.

Mark Allocation

Marks	Criteria
13-15	Clear explanation of the cause-and-effect chain from Learning & Growth through Process and Customer to Financial; specific application to Bright Ltd's strategy (international expansion); explanation of how strategy formulation translates into BSC objectives; discussion of leading vs lagging indicators or short-termism prevention.
10-12	Correct cause-and-effect linkage explained with reference to the four perspectives; some application to Bright Ltd; strategy-execution mechanism described.
6-9	Some understanding of the BSC's strategic role; linkage between perspectives mentioned but not fully developed; limited application to Bright Ltd.
2-5	General statement that BSC links strategy to performance without explanation of the mechanism; limited understanding of the cause-and-effect chain.
0-1	No meaningful understanding demonstrated.

Question 3 · Part 2**15 marks**

“Standard Costs are a detailed financial expression of organisational objectives.” Describe THREE non-financial objectives that organisations might have. In your answer, identify the stakeholder group corresponding with each non-financial objective.

Full Model Answer**1. Customer Service Quality — Stakeholder: Customers**

A non-financial objective of achieving and maintaining a customer satisfaction score of 90% or above (measured through post-interaction surveys or Net Promoter Score). Customers are the primary stakeholder group for this objective: they directly experience the quality of service and their loyalty to the organisation depends on consistently positive interactions. While financial standards measure the cost of service delivery, this non-financial objective captures the *outcome* from the customer’s perspective – something no financial metric alone can measure. Organisations that systematically track customer satisfaction and set improvement targets tend to demonstrate lower churn and higher lifetime customer value over time.

2. Employee Well-Being and Staff Retention — Stakeholder: Employees

A non-financial objective of reducing annual employee turnover to below 10% and achieving a minimum employee engagement score of 75% in the annual staff survey. Employees are the stakeholder group – they are directly affected by working conditions, management quality, development opportunities, and organisational culture. High turnover imposes recruitment and training costs that are financially significant, but more fundamentally it disrupts operational continuity and erodes institutional knowledge. Setting measurable non-financial targets for employee well-being signals organisational commitment to people beyond what financial budgets alone convey.

3. Environmental Impact Reduction — Stakeholder: Community / Society

A non-financial objective of reducing carbon emissions per unit of output by 15% over a defined period, or achieving certification under a recognised environmental standard (e.g. ISO 14001). The primary stakeholder group is the wider community and society, who bear the environmental consequences of an organisation’s operations even when they are not direct customers or employees. Setting non-financial environmental targets reflects the organisation’s accountability to this stakeholder group and is increasingly expected by regulators, investors, and the public as part of a credible sustainability strategy.

Mark Allocation

Marks	Criteria
13-15	Three distinct, well-developed non-financial objectives; each clearly non-financial in nature (not a reworded financial target); correct and clearly justified stakeholder group for each; objectives explained with sufficient depth (not just a label – the mechanism, measurement, and stakeholder relevance are articulated).
10-12	Three objectives identified with stakeholders; objectives are genuinely non-financial; one or two lack full development or stakeholder justification.
6-9	Three objectives attempted but one or two are essentially financial objectives reworded; stakeholders identified for most; some depth of explanation.
2-5	One or two valid non-financial objectives; stakeholders partially identified; limited explanation.
0-1	No meaningful attempt, or objectives are financial rather than non-financial.

QUESTION 3 TOTAL — 50 MARKS